

CLASS 10 · CBSE · NCERT MATHEMATICS · FORMULA BANK

Chapter 11: Areas Related to Circles

Every formula and result of the chapter on one sheet

1	Circumference of a circle	$2\pi r$
2	Area of a circle	πr^2
3	Area of a sector of angle θ	$\frac{\theta}{360} \times \pi r^2$
4	Length of an arc of angle θ	$\frac{\theta}{360} \times 2\pi r$
5	Area of a sector from its arc length l	$\frac{1}{2} l r$
6	Perimeter of a sector	$\frac{\theta}{360} \times 2\pi r + 2r$
7	Area of the major sector	$\pi r^2 - \text{area of the minor sector}$
8	Area of a minor segment	$\text{area of sector} - \text{area of triangle}$
9	Area of the major segment	$\pi r^2 - \text{area of the minor segment}$
10	Triangle in a sector, $\theta = 60^\circ$	$\frac{\sqrt{3}}{4} r^2$
11	Triangle in a sector, $\theta = 90^\circ$	$\frac{1}{2} r^2$
12	Triangle in a sector, $\theta = 120^\circ$	$\frac{\sqrt{3}}{4} r^2$
13	Area of a ring	$\pi(R^2 - r^2)$
14	Angle turned by the minute hand	6° per minute
15	Angle turned by the hour hand	30° per hour, 0.5° per minute